Customer Shopping Behavior Analysis

Data Analytics Project Report

Tools & Technologies: *Python, Pandas, MySQL, SQL, Power BI, VS Code*

1. Project Overview

This project analyzes customer shopping behavior using a retail shopping dataset containing 3,900 records and 18 columns.

The objective is to understand purchasing patterns, product category preferences, customer demographics, payment preferences, discount usage, and seasonal spending.

The project uses Python for data cleaning and exploratory data analysis, MySQL for business analysis queries, and Power BI to create an interactive dashboard.

2. Problem Statement

A retail company wants to better understand customer shopping behavior to improve sales, customer satisfaction, and long-term loyalty.

Management wants to identify trends in purchasing patterns across demographics, product categories, discounts, reviews, seasons, and payment preferences.

The objective is to use shopping data to identify meaningful insights and support better marketing and product-related decisions.

3. Project Objectives

- Clean and prepare the raw shopping dataset.
- Perform exploratory data analysis using Python.
- Store and analyze structured data using MySQL.
- Write SQL queries to answer business questions.
- Develop an interactive Power BI dashboard.
- Identify important shopping trends and provide business recommendations.

4. Dataset Description

The dataset contains:

Metric	Value
Total records	3,900
Total columns	18
Unique Customer IDs	3,900
Missing values before cleaning	37
Duplicate rows	0

Important columns include Age, Gender, Category, Item Purchased, Purchase Amount, Season, Review Rating, Subscription Status, Discount Applied, Payment Method and Frequency of Purchases.

5. Data Cleaning Using Python

Python and Pandas were used to inspect and clean the dataset.

The following steps were performed:

1. Loaded the raw CSV dataset using Pandas.
2. Checked dataset dimensions, column names and data types.
3. Identified missing values.
4. Checked duplicate records.
5. Examined numerical statistics and categorical values.
6. Handled missing Review Rating values.
7. Exported the cleaned dataset.

Missing value treatment:

There were 37 missing values in the Review Rating column. These were replaced with the column mean of approximately 3.75.

After cleaning, the dataset contained 3,900 records with no missing values or duplicate rows.

The cleaned dataset was saved as:

customer_shopping_behavior_cleaned.csv

6. Exploratory Data Analysis (EDA)

EDA was performed to understand customer purchasing patterns.

Major analyses included:

- Purchase count by product category.
- Average purchase amount by category.
- Spending comparison by gender.
- Subscription status analysis.
- Discount usage analysis.
- Payment method distribution.
- Purchase frequency analysis.
- Product category distribution by season.
- Seasonal purchase analysis.

Key Python Findings

Product Category Distribution

Category	Purchase
Clothing	1,737
Accessories	1,240
Footwear	599
Outerwear	324

Clothing was the most frequently purchased category.

Average Purchase Amount by Category

Category	Average (USD)
Footwear	60.26
Clothing	60.03
Accessories	59.84
Outerwear	57.17

Footwear recorded the highest average purchase amount.

Payment Preferences

PayPal was the most frequently used payment method, with 677 records (17.36%).

7. SQL Business Analysis Using MySQL

The cleaned dataset was imported into the customer_shopping_analysis database in MySQL.

A table named customer_shopping was created, and 3,900 records were successfully inserted.

SQL was used to perform business analysis, including:

- Verifying total records and unique customer IDs.
- Calculating total and average purchase amounts.
- Analyzing product category performance.
- Comparing purchasing patterns by gender.
- Analyzing discount usage.
- Examining payment preferences.
- Analyzing seasonal purchasing behavior.
- Comparing customer review ratings by category.

These queries helped validate the Python findings and answer business-related questions.

8. Power BI Dashboard

Power BI Desktop was used to create an interactive dashboard using the cleaned dataset.

The dashboard contains two pages.

Page 1: Customer Overview

This page includes:

- Total purchase amount.
- Total customer records.
- Average purchase amount.
- Average review rating.
- Purchase count by category.
- Average purchase amount by category.
- Customer age-group distribution.
- Average purchase amount by gender.

Interactive slicers were added for Category, Gender and Season.

Page 2: Shopping Preferences

This page includes:

- Discount usage distribution.
- Payment method distribution.
- Average purchase amount by season.
- Subscription status distribution.

Interactive slicers were added for Discount Applied and Subscription Status.

The slicers allow users to explore the data dynamically.

9. Key Business Insights

1. Clothing represents the largest product category, with 1,737 records.
2. Footwear has the highest average purchase amount at $60.26.
3. PayPal is the most frequently recorded payment method.
4. The average purchase amount across all records is approximately $59.76.
5. Fall has the highest average purchase amount among the four seasons, at $61.56.
6. Records without discounts have a slightly higher average purchase amount ($60.13) than records with discounts ($59.28).
7. The average review rating is approximately 3.75.

10. Business Recommendations

Based on the analysis:

- **Product strategy:** Review the strong purchase volume in Clothing and explore opportunities to improve performance in Outerwear.
- **Marketing strategy:** Use category and seasonal purchasing patterns to plan targeted promotions.
- **Payment experience:** Continue supporting popular payment methods, including PayPal and credit cards.
- **Discount strategy:** Evaluate discount effectiveness instead of assuming that more discounts always increase spending.
- **Customer experience:** Monitor review ratings and identify opportunities to improve customer satisfaction.

11. Limitations

- The dataset contains 3,900 unique Customer IDs, but does not provide multiple transaction records per customer.
- The dataset does not contain a sales channel column, so online versus offline purchasing cannot be compared.
- The analysis is descriptive and does not establish that discounts or other factors cause changes in spending.
- The 37 missing review ratings were replaced with the mean, which may influence rating-related analysis.

12. Conclusion

This project demonstrates an end-to-end data analytics workflow, starting from raw data inspection and cleaning to exploratory analysis, SQL business queries and interactive Power BI visualization.

Python helped prepare and explore the dataset, MySQL enabled structured business analysis, and Power BI presented the findings through an interactive dashboard.

The project provided insights into product category preferences, spending patterns, payment methods, discount usage and seasonal behavior.

It demonstrates practical application of data analytics tools and techniques to support data-driven business decisions.

Project Status: Completed

Technologies: Python | Pandas | MySQL | SQL | Power BI